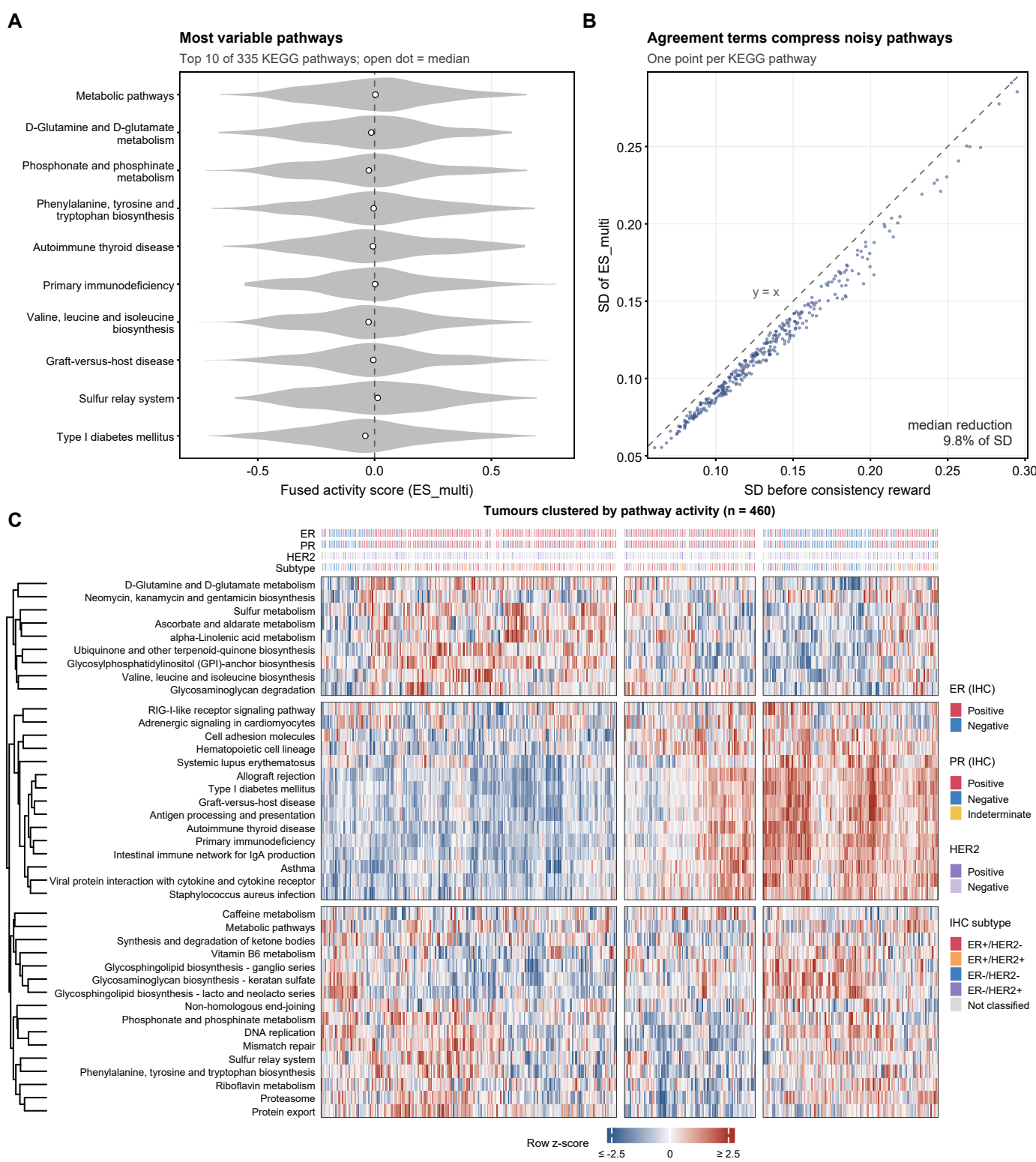

A
Most variable pathways
Top 10 of 335 KEGG pathways; open dot = median
Metabolic pathways
D-Glutamine and D-glutamate metabolism
Phosphonate and phosphinate metabolism
Phenylalanine, tyrosine and tryptophan biosynthesis
Autoimmune thyroid disease
Primary immunodeficiency
Valine, leucine and isoleucine biosynthesis
Graft-versus-host disease
Sulfur relay system
Type I diabetes mellitus
-0.5
0.0
0.5
Fused activity score (ES_multi)
B
Agreement terms compress noisy pathways
One point per KEGG pathway
SD of ES_multi
0.05
0.10
0.15
0.20
0.25
y = x
median reduction
9.8% of SD
0.10
0.15
0.20
0.25
0.30
SD before consistency reward
C
Tumours clustered by pathway activity (n = 460)
ER
PR
HER2
Subtype
D-Glutamine and D-glutamate metabolism
Neomycin, kanamycin and gentamicin biosynthesis
Sulfur metabolism
Ascorbate and aldarate metabolism
alpha-Linolenic acid metabolism
Ubiquinone and other terpenoid-quinone biosynthesis
Glycosylphosphatidylinositol (GPI)-anchor biosynthesis
Valine, leucine and isoleucine biosynthesis
Glycosaminoglycan degradation
RIG-I-like receptor signaling pathway
Adrenergic signaling in cardiomyocytes
Cell adhesion molecules
Hematopoietic cell lineage
Systemic lupus erythematosus
Allograft rejection
Type I diabetes mellitus
Graft-versus-host disease
Antigen processing and presentation
Autoimmune thyroid disease
Primary immunodeficiency
Intestinal immune network for IgA production
Asthma
Viral protein interaction with cytokine and cytokine receptor
Staphylococcus aureus infection
Caffeine metabolism
Metabolic pathways
Synthesis and degradation of ketone bodies
Vitamin B6 metabolism
Glycosphingolipid biosynthesis - ganglio series
Glycosaminoglycan biosynthesis - keratan sulfate
Glycosphingolipid biosynthesis - lacto and neolacto series
Non-homologous end-joining
Phosphonate and phosphinate metabolism
DNA replication
Mismatch repair
Sulfur relay system
Phenylalanine, tyrosine and tryptophan biosynthesis
Riboflavin metabolism
Proteasome
Protein export
ER (IHC)
Positive
Negative
PR (IHC)
Positive
Negative
Indeterminate
HER2
Positive
Negative
IHC subtype
ER+/HER2-
ER+/HER2+
ER-/HER2-
ER-/HER2+
Not classified
Row z-score
≤ -2.5
0
≥ 2.5